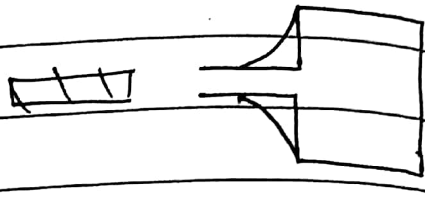
2/1/18.

**Hot Tear**

Due to high residual stresses.



**Shrinkage Cavity.**

Improper riser.

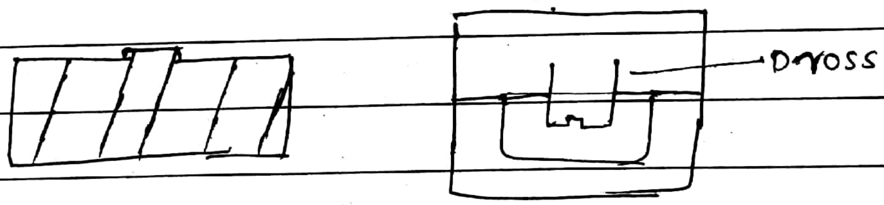
**Shift.**

Misalignment b/w Cope and drag or misalignment of Core.

**Honey Comb.**

Defect shaped like honey comb, due to dirt/impurities mechanically held in suspension in molten metal.

**Dross.**

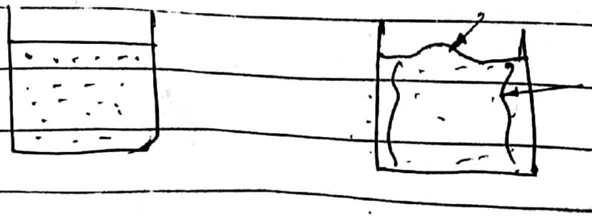



Happens due to the sand removal.

**Dirt**

**Wash**

**Buckle.**



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# PRODUCT DESIGN CONSIDERATION FOR DEFECT FREE CASTING

1. Shrinkage – Making Pattern oversize.

Date / /

# DESIGN FOR ECONOMICAL CASTING

{Product design for economical Sand casting}

optimum Riser Design.
optimum Gating Design.
Mould Filling Simulation
Sand Parameters optimisation.
Deformation control.
Efficient cooling methods.

chvorinov's Rule.

$$\text{Solidification time} = K\left(\frac{V}{SA}\right)^2.$$

K = constant,

V = Volume of casting

S.A = Surface Area.

For directional solidification

$$\left(\frac{V}{SA}\right)_{riser} > \left(\frac{V}{SA}\right)_{casting}$$

Caine's Formulae.

$$\text{Freezing ratio } X = \frac{(S.A/V)_{casting}}{(SA/V)_{riser}}$$

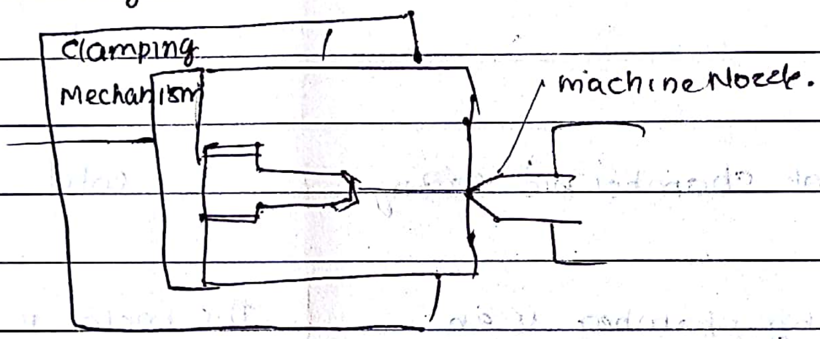
Volume ratio : Y = Vol of riser / Vol of casting.

$$\text{Caines Relation } X = \frac{a}{y-b} + c -$$

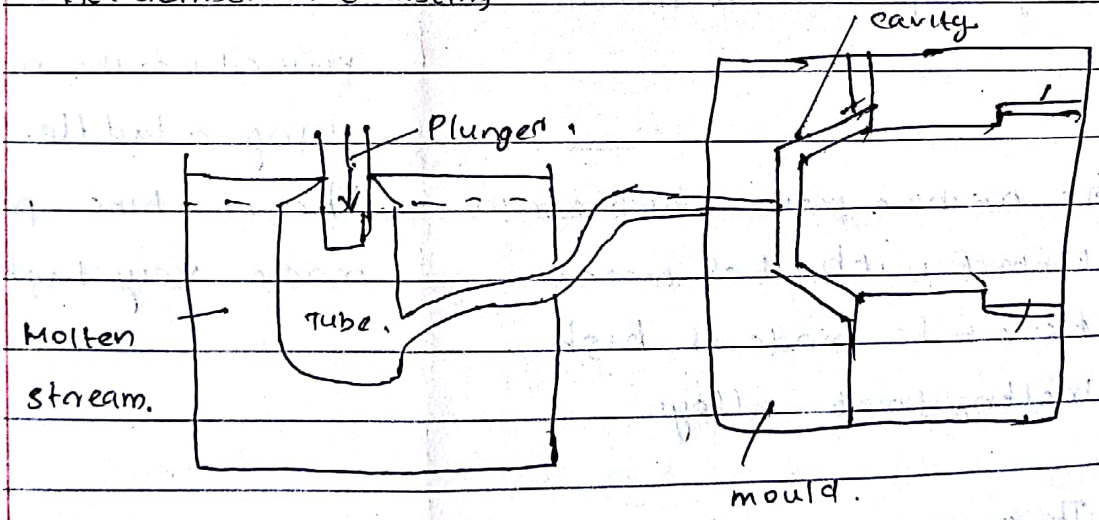
a = freezing constant
b = Liquid - Solid solidification contraction coeff.
c = Relative freezing rate of mould casting

assignment
which are diff type of sand moulding machines / equipments.

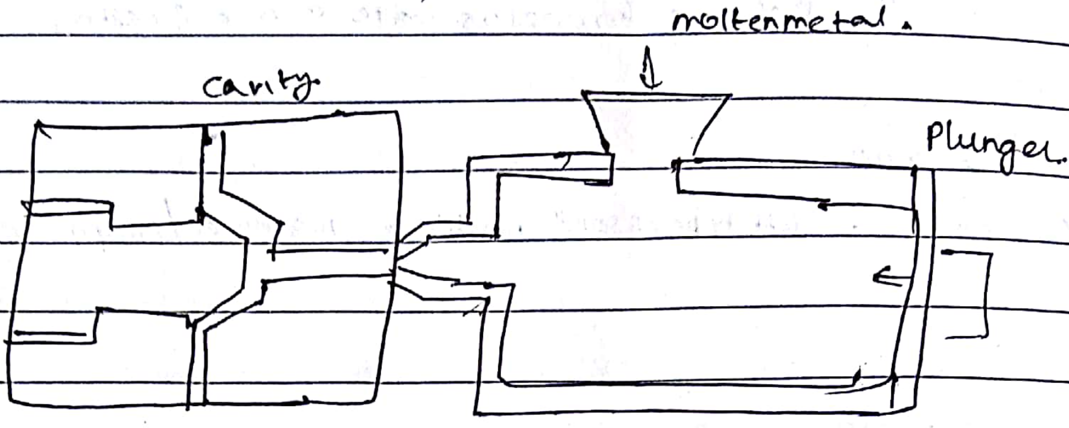
Die casting - Permanent -




Hot chamber - Die casting



Cold chamber die casting.




Hot chamber die casting — cold

| Hot chamber die casting | cold |
|---|---|
| The melting chamber is an integral part of the machine. | The metal is melted in a seperate furnace! And is then poured into the machine by using a laddle. |
| The machine parts which are in contacts with hot metals have to be made of high melting point alloys. | The machine parts doesn't made very high tempalloys. |
| The pressure used for casting is lower | The pressure used for casting are higher |
| Castings are less denser and may have porosity | Castings are more dense and is free from any porosity defect |
| The metals casted by thi methods are zink, aluminium or their alloys | The metal casted by thi' method are magnesium, copper & its alloys |

casting size is smaller but the production rate is faster

Casting size is larger but production rate is smaller.

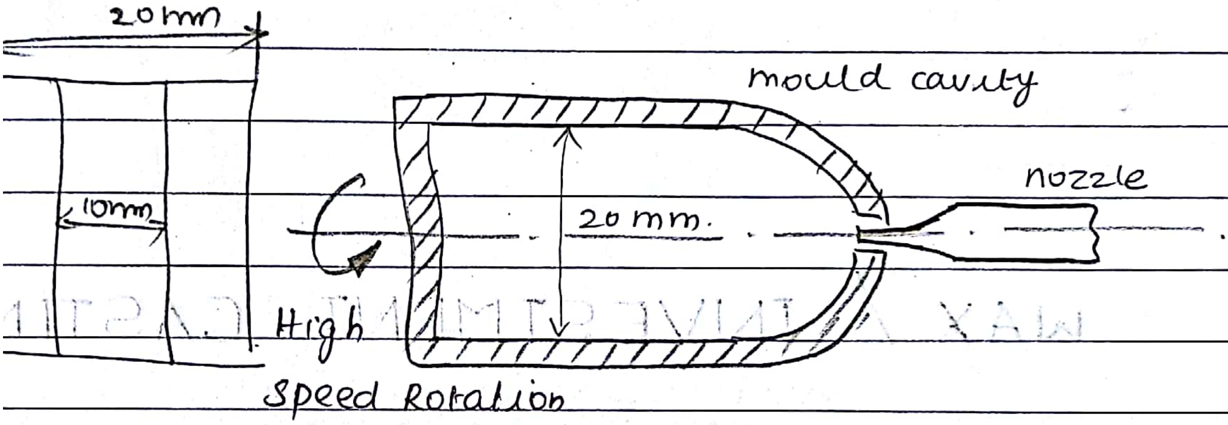
Centrifugal Casting

→ True Centrifugal casting ⇒ Vertical & Horizontal.

→ Semi " ⇒ No impurities, Denser

→ Centrifuging ⇒ Independent Cavities

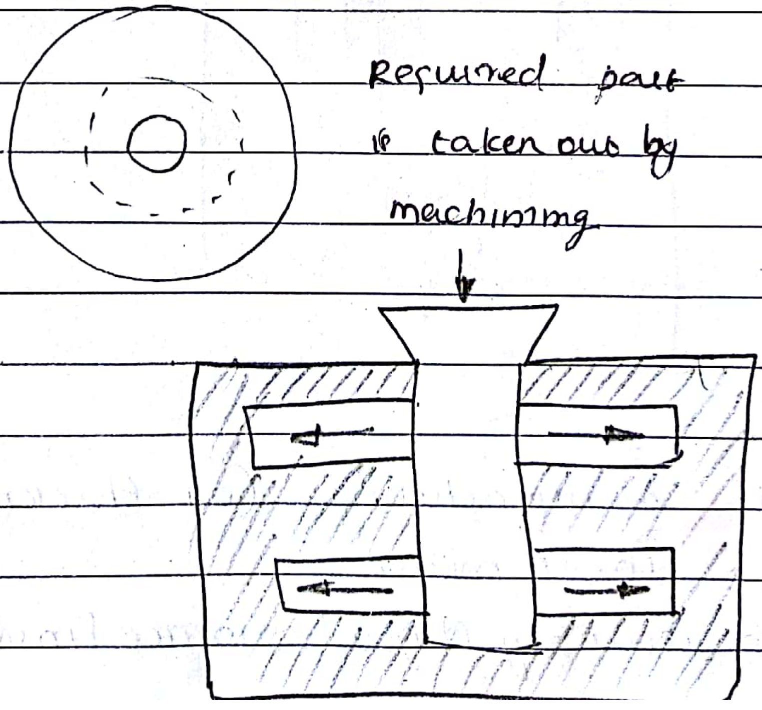
→ Centrifugal




→ Semi centrifugal

purely ~~[illegible]~~ impurified metal

more solid.



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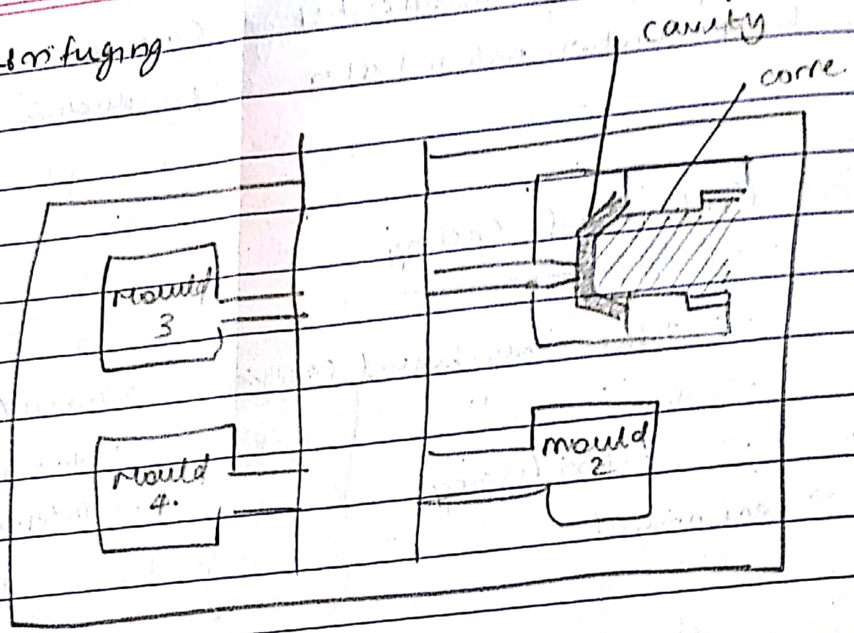
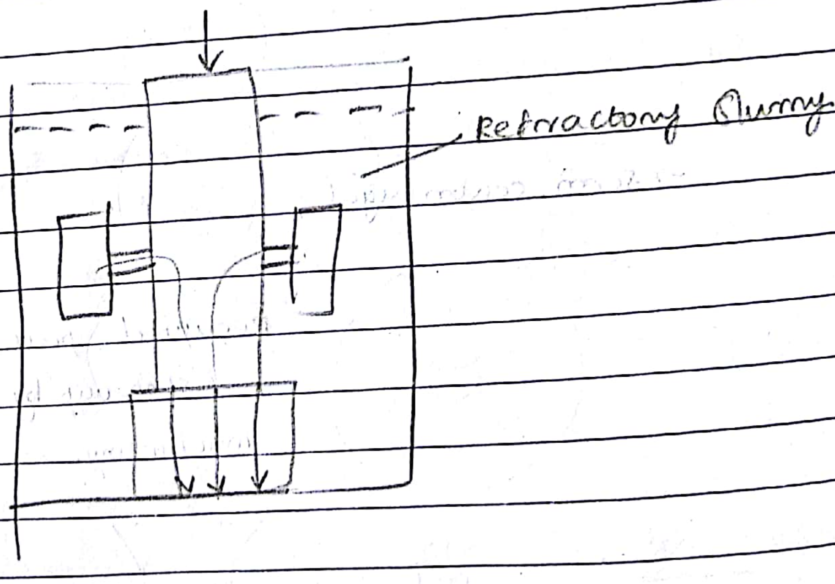
↳ Centrifuging




# WAX / INVESTMENT CASTING

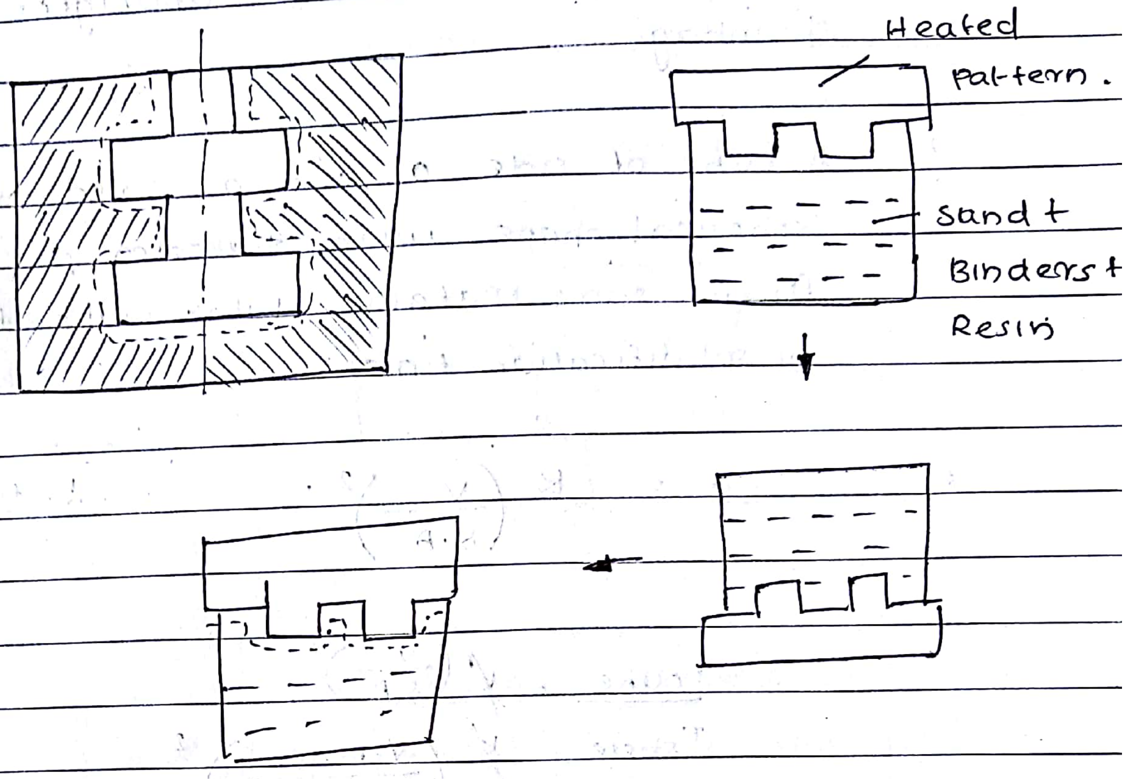



Expensive
Fine parts manufacturing (0.1 thickness)
accuracy 10-20 micron
High dim accuracy & good surface finish.

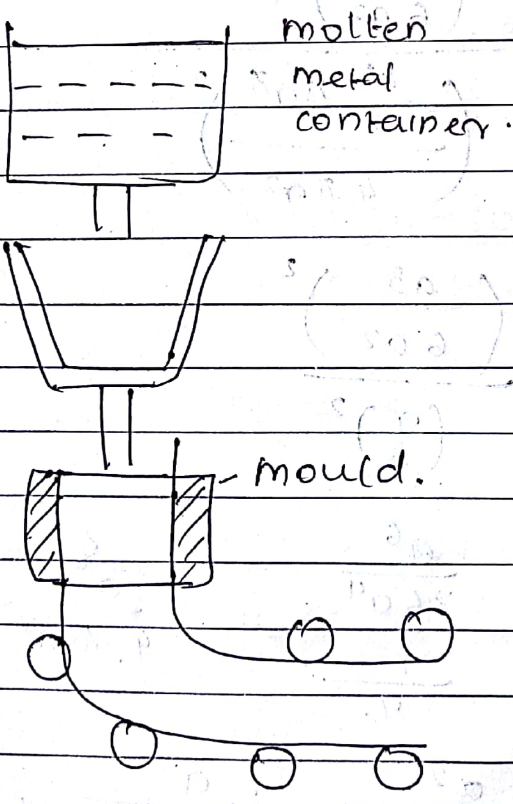
Scanned by CamScanner

31-1-2018

# SHELL MOULDING PROCESS




# CONTINOUS CASTING



During the cooling of a casting having cubical shape with side 40 mm undergoes a volume shrinkage

A cube of side a and a solid which is having spherical shape with diameter 'd' are made from same material. What will be their ratio of solidification time.

$$t = K\left(\frac{V}{S \cdot A}\right)^2 \qquad t = \text{time}$$

$$\frac{T_{cube}}{T_{sphere}} = \frac{\cancel{K}\left(\frac{V}{SA}\right)^2}{\cancel{K}\left(\frac{V_{sphere}}{(SA)_{sphere}}\right)^2}$$

$$= \frac{\left(\frac{a^3}{6a^2}\right)^2}{\left(\frac{\frac{\cancel{4}}{3}\cancel{\pi} r^{\cancel{3}}}{\cancel{4}\cancel{\pi} \cancel{r^2}}\right)^2}$$

$$= \frac{\left(\frac{a^3}{6a^2}\right)^2}{\left(\frac{r}{3}\right)^2}$$

$$= \frac{\frac{a^6}{36a^4}}{\frac{r^2}{9}} = \frac{a^{\cancel{6}\,2}}{4 \cdot 36 a^{\cancel{4}}} \times \frac{9}{r^2}$$

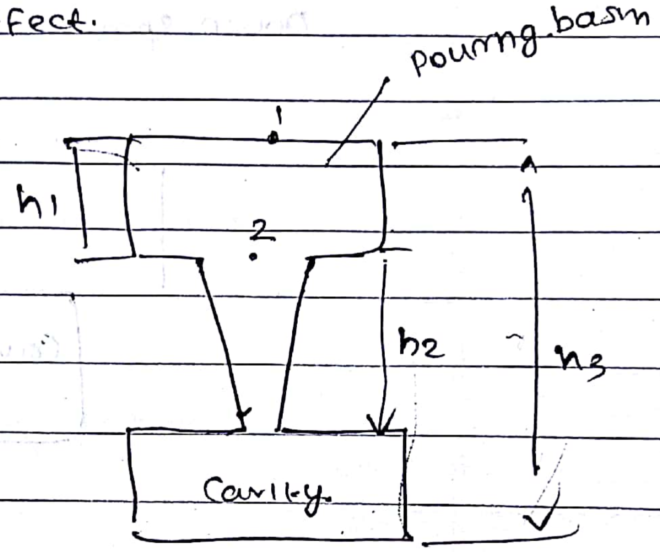
$$= \frac{a^2}{4r^2} = \frac{a^2}{4\left(\frac{d}{2}\right)^2} = \frac{a^2}{\cancel{4} \cdot \frac{d^2}{\cancel{4}}}$$

$$= \left(\frac{a}{d}\right)^2$$

SPRUE DESIGN

To avoid aspiration effect.

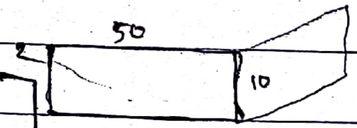
$$R = \frac{A_3}{A_2} = \sqrt{\frac{h_1}{h_3}}$$




? A casting of size 50 cm x 40 cm x 10 cm solidify in 20 min what will be the solidification time for an another similar casting having a size 40 x 30 x 5 cm.

A.

$$20 \propto \left[\frac{50 \times 40 \times 10}{(50 \times 40 + 50 \times 10 + 40 \times 10)}\right]^2$$




$$t \propto \left[\frac{40 \times 30 \times 5}{(40 \times 30 + 40 \times 5 + 30 \times 5)}\right]^2$$

$$\frac{20}{t} = \frac{11.89}{3.74}$$

$$20 \times 3.74 = t \times 11.89$$

$$t = \frac{20 \times 3.74}{11.89} = 6.29 \text{ min}$$

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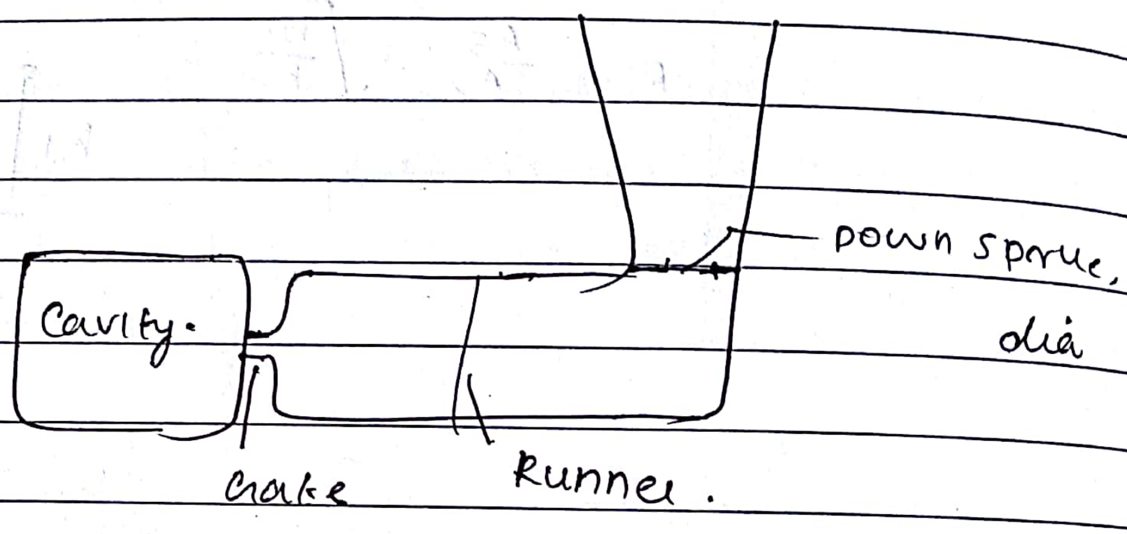
GATING RATIO

Down sprue : Runner : Gate.




For pressurised casting = 2:2:1
Non pressurised casting : (Eg : sand casting)

Ratio = 1:2:2 or 1:4:4

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14/2/2018 CASTING CLEANING

1. Surface cleaning.
2. Trimming
3. Inspection → Destructive Test
   → Non destructive test.

Assignment
Sand casting instruments & Machines.

Sand blasting.
Tumbling.

Modulus Method of Riser.
↓
Chvorinov's Rule, $\left(\frac{SA}{V}\right)_{Riser} < \left(\frac{SA}{V}\right)_{casting}$.